Wenbo Lyu

B.Sc. Graduate

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Curriculum Vitae
October 2026

Education and qualifications

B.Sc. in GIScience Shaanxi Normal University 2025

Research Experience

2026.6- Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6 Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8 Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- **Goal:** Understanding complex systems via observational data (temporal/spatial/spatiotemporal data).
- **Methods:** *Information theory*, *dynamic systems*, *persisent homology*, *information geometry*, *optimal transport*, and *generative models* for *spatial causal inference*.
- **Tech Stack:** *R, C++, Python* (spatial analysis & open-source geospatial software).

Selected research papers

1. Lyu, W, Qin, Y, Zhang, P, Yi, W, Song, Y, Li, X, Dai, S, Yuan, T, Zheng, C, Stein, A, and Zhao, W (2026). Spatially convergent partial cross mapping for identifying direct causal influence in cross-sectional data. *Communications Physics 9*(1). DOI: 10.1038/s42005-025-02528-1.
2. Lyu, W, Dai, S, Song, Y, Zhao, W, Yi, W, Xiao, Y, and Jia, N (2026). Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality. *International Journal of Geographical Information Science*, 1–23. DOI: 10.1080/13658816.2026.2687121.
3. Lyu, W, Lei, Y, Yi, W, Song, Y, Li, X, Dai, S, Qin, Y, and Zhao, W (2026). Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM. *Computers, Environment and Urban Systems 127*, 102435. DOI: 10.1016/j.compenvurbsys.2026.102435.
4. Lv, W, Lei, Y, Liu, F, Yan, J, Song, Y, and Zhao, W (2025). gdverse: An R Package for Spatial Stratified Heterogeneity Family. *Transactions in GIS 29*(2), e70032. DOI: 10.1111/tgis.70032.
5. Lv, W, Liu, F, Cai, K, Cao, Y, Deng, M, Liang, W, Yan, J, and Wang, G (2024). Distinguishing the Impacts and Gradient Effects of Climate Change and Human Activities on Vegetation Cover in the Weihe River Basin, China. *Journal of Geophysical Research: Biogeosciences 129*(10), e2024JG008297. DOI: 10.1029/2024jg008297.
6. Lei, Y, Lyu, W, Zhang, P, Xu, D, Zhang, Z, Li, X, Song, Y, Qin, Y, Yuan, T, and Zhao, W (2026). Spatial Pattern Entropy: A Symbolic Measure for Continuous Spatial Data. *International Journal of Geographical Information Science*, 1–23. DOI: 10.1080/13658816.2026.2757205.
7. Li, B, Lv, W, Ding, H, Song, Y, and Zhao, W (2026). New-type urbanization policy, urban expansion, and employment quality in chinese cities: evidence from a quasi-natural experiment in China. *GIScience & Remote Sensing 63*(1), 2716413. DOI: 10.1080/15481603.2026.2716413.
8. Xu, L, Shi, H, Lyu, W, Zhang, Y, and Zhao, H (2026). Beyond correlation: Unveiling the causal link between street-scapes and recreational walking/cycling. *Journal of Transport Geography 136*, 104789. DOI: 10.1016/j.jtrangeo.2026.104789.
9. Chen, C, Song, Y, Lv, W, Shemery, A, Hampson, K, Yi, W, Zhong, Y, and Wu, P (2025). Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning. *Computer-Aided Civil and Infrastructure Engineering 40*(20), 3074–3092. DOI: 10.1111/mice.13489.

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.

Unpublished working papers

Quantifying pairwise directional influence between spatial variables: A reconstruction-based approach to asymmetric relationships
Unveiling interactions between spatial processes with information decomposition
Enhancing spatial stratified heterogeneity measurement for multivariate continuous data via distance ranks
Pattern Causality for Unveiling Interactions with the R Package pc
coupling: Coupling Coordination Analysis in R
Spatial causal discovery under latent confounders via deep orthogonal representation learning

Selected awards and honours

2024 Longi Non-Education Major Scholarship
2024 First Prize in the 13th National University Student GIS Application Skills Competition
2023 Grand Prize in the 12th National University Student GIS Application Skills Competition
2023 First Prize in the Second National University Student Ecological Environment Management Research Innovation Competition
2023 Second Prize of the 5th 'Guodi Cup' National College Student Natural Resource Science and Technology Competition
2021 Outstanding Individual in Military Training Publicity for College Students, Shaanxi Normal University

Software (R packages)

`infoxtr`	Information-Theoretic Measures for Revealing Variable Interactions
`spEDM`	Spatial Empirical Dynamic Modeling
`tEDM`	Temporal Empirical Dynamic Modeling
`pc`	Pattern Causality Analysis
`gdverse`	Analysis of Spatial Stratified Heterogeneity
`coupling`	Coupling Coordination Analysis
`sdsfun`	Spatial Data Science Complementary Features